

AUTOSMART PARTS

PHASE 2: Multi-Agent AI System

TEAM 1
Phase 1

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FROM PHASE 1 to PHASE 2

02/10

Phase 1



Basic AI Chatbot

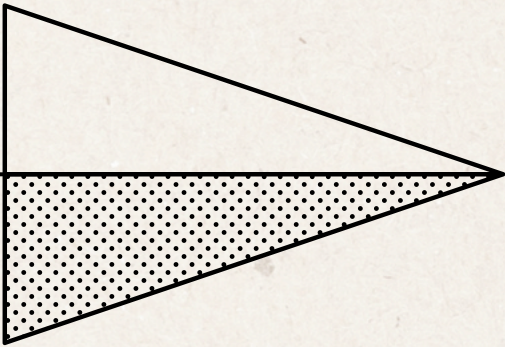
Product recommendations
Simple e-commerce platform

Phase 2:



Multi-Agent Ecosystem

Specialized AI agents
Computer vision integration
Automated decision engine
Complete analytics



5 SPECIALIZED AI AGENTS

Recommendation Agent

Conversational product matching

Fraud Detection Agent

Analyzes suspicious transactions

Vision API for document analysis

Defect Detection Agent

Identifies product defects via images

Damage Assessment Agent

Evaluates shipping damage from photos

AI Decision Engine

Confidence-based actions

Automated REFUND/REPLACE/
ESCALATE

TECHNOLOGY ENHANCEMENTS



Frontend	React + File Upload Components
Backend	Node.js + JWT Authentication
Database	PostgreSQL (18 tables)
Open AI	OpenAI GPT-4o-mini + Vision API
Storage	Multer + S3 for image uploads
Analytics	Comprehensive event tracking

PHASE 2 ACHIEVEMENTS

Computer Vision Integration

Image analysis for defects/
damage/fraud

10MB file uploads with preview

Automated Decision Engine

Confidence scoring (0-100%)

Auto-refund/replacement triggers

Human escalation for edge cases

Complete Order Management

Full order lifecycle tracking

User authentication & history

Real-time status updates

DATA INFRASTRUCTURE

18 Database Tables

Users, Orders, Products

Support Tickets, AI
Interactions

Analytics Events, Audit Logs

Performance Tracking

AI interactions logged

Purchase conversion rates

Confidence score analytics

Resolution time monitoring

System Performance Metrics

07/10

Accuracy	100% across all agents
Confidence	86.75% average score
Conversion	56.8% purchase rate from AI
Vision Analysis	2-4 second processing
Decisions	REFUND/REPLACE/ESCALATE with audit trail

Technical Challenges Overcome

08/10

Database Integration	PostgreSQL with 18 tables
Multi-Agent Coordination	Specialized routing
Computer Vision	OpenAI Vision API integration
File Management	Secure upload & storage
Decision Automation	Confidence-based workflows

CUSTOMER JOURNEY / LIVE DEMO

- 1. Browse & Get AI Recommendations
- 2. Purchase with Secure Checkout
- 3. Track Orders & Shipping Status
- 4. Submit Support Tickets with Images
- 5. AI Analyzes & Makes Decisions
- 6. Automated Resolution Execution
- 7. Analytics Track Performance

DELIVERABLES MET

- Multi-Agent AI System
- Computer Vision Capabilities
- Automated Decision Engine
- Comprehensive Analytics
- Enterprise-Grade Infrastructure

Demo